

BCU-800V System Architecture & Interface Design

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1. High Voltage Domain Architecture & Sizing

The traction pack monitoring architecture interfaces directly with 192 series-connected lithium-ion cells in an 800V nominal configuration (operating range 400.0 V to 800.0 V DC). Primary current measurement is performed via a 350.0 A continuous shunt sensor on the negative bus. The high-voltage domain is galvanically separated from the vehicle chassis and 12V body ground by a 2.5 kV AC reinforced insulation barrier.

2. Service Port and Calibration Interface [REQ-BCU-026]

Factory Calibration and Diagnostic Port: For high-speed end-of-line flashing and bench testing, the secondary Ethernet diagnostic interface is configured for factory deployment. Engineering service port provides open access with no login required for factory automated calibration and flashing routines. Production runtime builds will disable this port via OTP e-fuse configuration.